

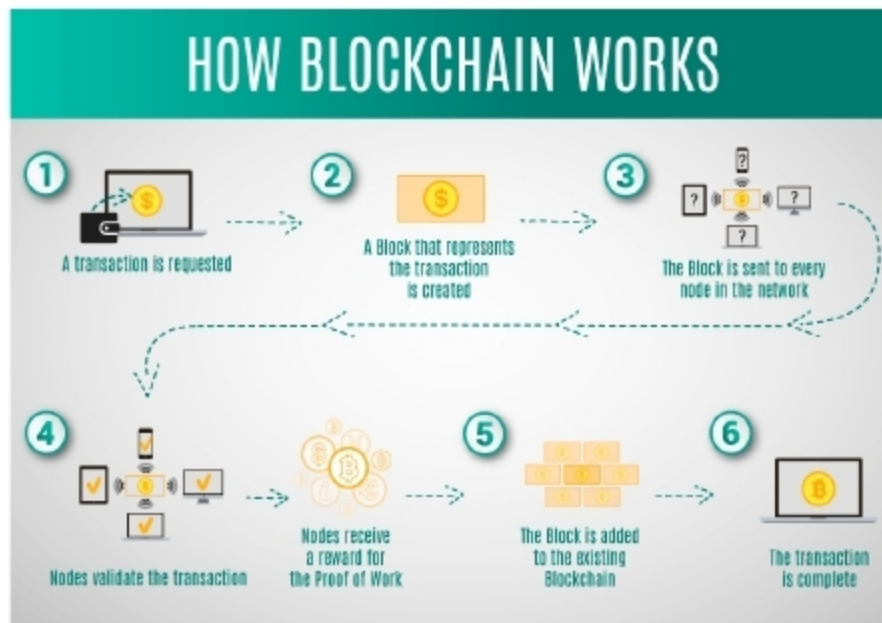


Figure 1: How the blockchain works (Image source: [googleimages.com](#))

centralised systems do have a few vulnerabilities associated with them, like being easy targets for hackers, difficult to upgrade, no proper backup if the system shuts down, etc. Considering all this, the concept of decentralised systems was introduced. In such systems, the information is not stored by a single entity. Everyone in the network is the owner of the information. If the first party of the system wants to interact with the second one, both can directly interact with each other without the intermediary third centralised entity. In the case of Bitcoins, we can send our money to anyone on the chain without having to go through a bank.

Transparency: One of the most misunderstood concepts about blockchain technology is transparency. Some of us understand that a blockchain gives us privacy while some of us consider it as transparent. In a blockchain, everyone's identity is masked by complex cryptography and is represented by their public address only. So if we look at a person's transaction history in a blockchain, we will not see something like 'Bob sent US\$ 100' but instead, we will see something like `1MF1bhsFLkBzzz9vpFYEmvwT2TbyCt7NZJ sent $100`. So, while the person's real identity is not revealed, we can still look at all the transactions that person has done using his or her public address.

This level of transparency never existed before within any financial system. If we consider this from a cryptocurrency point of view, we can simply pop the public address of a user in an explorer and look at the list of all the transactions that the person has engaged in, provided we already know the public address. This is what forces different companies using this technology to be honest.

Immutability: Immutability means that once something is entered into the blockchain, it cannot be tampered with. The blockchain supports this property as it stores data using the cryptographic hash function. Hashing refers to the process of taking an input string (of any length) and giving out a fixed-length output string. In case of cryptocurrencies like Bitcoin, different transactions are taken as inputs, and then run through a specific hashing algorithm that generates outputs of fixed length.

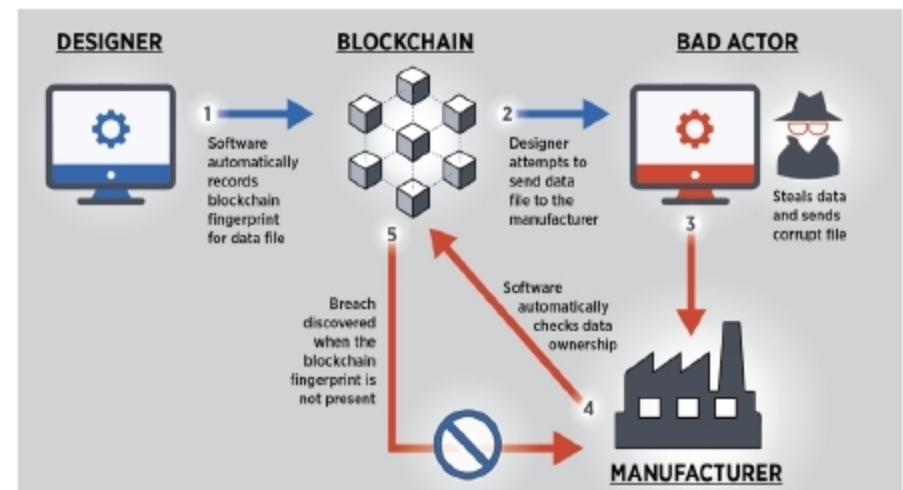


Figure 2: How the blockchain identifies a bad actor in the system (Image source: [googleimages.com](#))

This makes the blockchain so amazingly reliable. If a hacker attacks and tries to change the data of any specific block, then it will impact the data present in all previous blocks as all the blocks are interlinked through their addresses, hence changing the whole blockchain drastically.

This is what keeps the blockchain immutable.

Types of blockchain technologies

There are three different types of blockchain technologies.

Public blockchain: Public blockchain protocols are based on the Proof of Work (PoW) consensus algorithms and are available as open source. Anyone can participate in these without permission. Everyone can download the code and start running any of the public nodes on their local device to validate various transactions in the network. This process helps in determining the state of different blocks that get added to the chain. Anyone in any part of the world can send their transactions through this network, and can expect to see them included in the blockchain if they are valid. Also, anyone can read the transaction on a public block explorer. All the transactions are transparent but anonymous. This process doesn't incur much infrastructure cost. For example, Bitcoin, Ethereum, etc, fall under the public blockchain category.

Consortium blockchain: Consortium or federated blockchains always operate under the control of a group. In comparison to public blockchains, they do not allow just any person with Internet access to participate in the process of transaction verification. They are faster, more scalable and even provide more transaction privacy. Consortium blockchains are widely used in the banking sector. Here, the consensus process is in the control of a pre-selected set of nodes. The right to read the complete blockchain may be public, or may be restricted only to the pre-selected participants. This kind of blockchain reduces data redundancies and the associated transaction costs, hence simplifying the document handling process. It also helps in getting rid of the semi-manual compliance mechanisms. For example, EWF (from the energy sector) and R3 (in banking) are blockchains that fall under the consortium category.